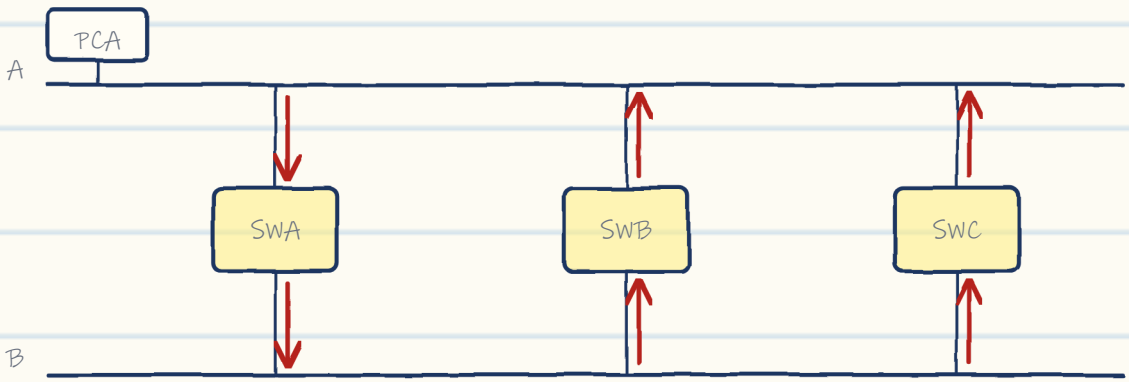
Qamrovchi daraxt protokollari

Spanning Tree Protocols

Asosiy atamalar
- Halqa (loop) — kadr aylanib qoladi, tarmoq bo'g'iladi
- STP — ortiqcha kanalni bloklab halqasiz daraxt yasaydi
- BPDU — kommutatorlar almashadigan boshqaruv kadri
- Root bridge — daraxt ildizi: eng kichik bridge ID
- Bridge ID — ustuvorlik (priority) + kommutator MAC manzili

1. Halqa nima qiladi?

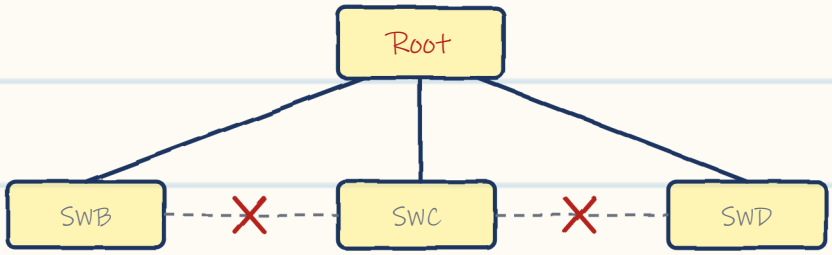


kadr ko'payadi va aylanadi — broadcast bo'roni

1-rasm. Ko'prik kadrni o'zgartirmaydi, hop hisoblagichi yo'q

- MAC jadval har safar buziladi — manzil sakraydi
- Har o'tishda kadrning nusxasi ikkiga ko'payadi

2. STP nima qiladi?

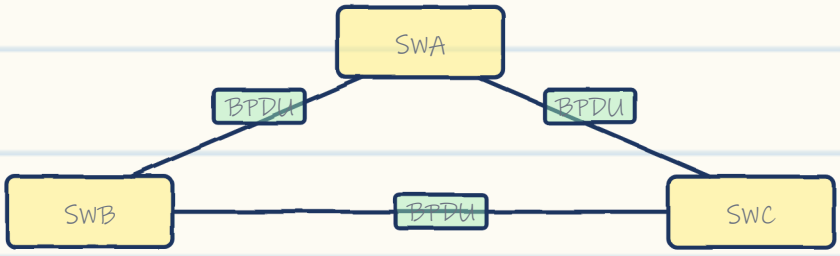


uzuq chiziq = bloklangan zaxira yo'l

faol yo'l uzilsa — zaxira ochiladi

2-rasm. 802.1D halqani daraxtga kesadi

3. BPDU

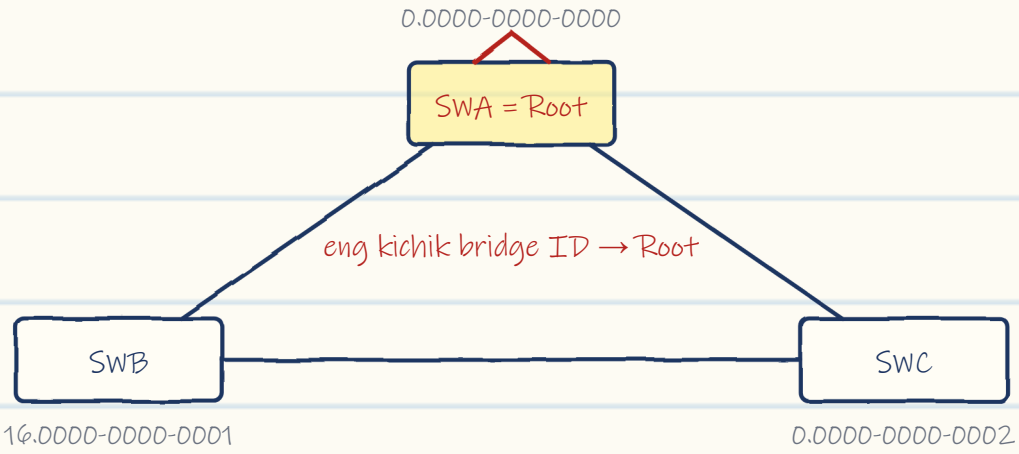


802.1D · daraxt BPDU almashinuvi bilan hisoblanadi

Root ID	Root path cost	Desig. bridge ID	Desig. port ID

- Root ID — jo'natuvchi root deb bilgan ko'prik ID'si
- Root path cost — root'gacha eng arzon yo'l narxi (root'da 0)
- Designated bridge ID — segment egasi (ustuvorlik + MAC)
- Designated port ID — port ustuvorligi + port indeksi

4. Root bridge tanlash

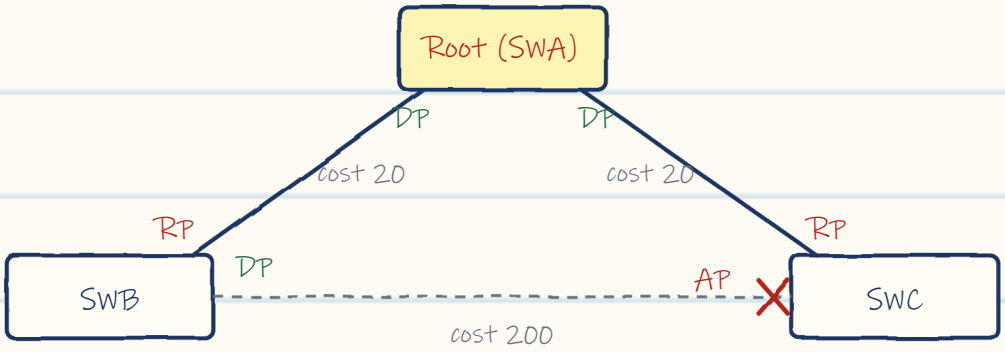


ustuvorlik (standart 32768) + MAC

3-rasm. Ustuvorlik teng bo'lsa — kichik MAC g'olib

Boshida har kommutator o'zini root deb biladi. Keyin BPDU'ni faqat root hosil qiladi, qolganlar uzatadi.

5. Port rollari



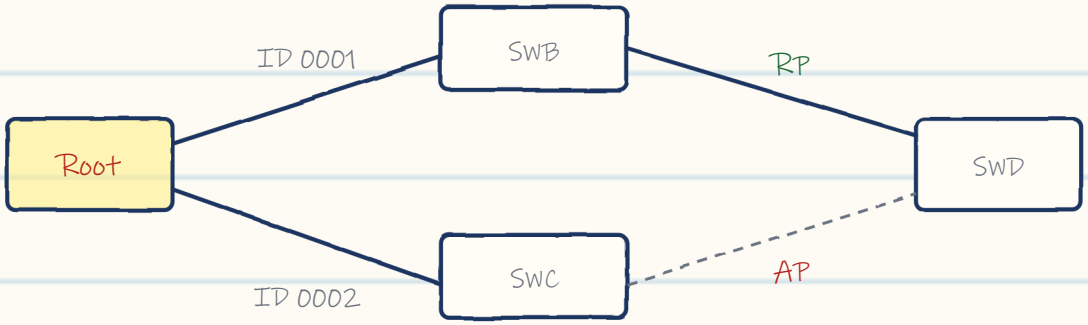
DP va RP uzatadi · AP bloklangan

root'ning hamma portlari — DP

- Root'da hamma port — DP (designated)
- Har root'siz ko'prikda eng arzon port — RP (root port)
- Har segmentda eng arzon ko'prik — designated bridge, uning porti — DP
- Qolgani — AP (alternate), data bloklanadi

6. Yo‘l narxi va teng holat

Tezlik	10 Mbit/s	100 Mbit/s	1 Gbit/s	10 Gbit/s
802.1D-1998	100	19	4	2
H3C (standart)	2000	200	20	2



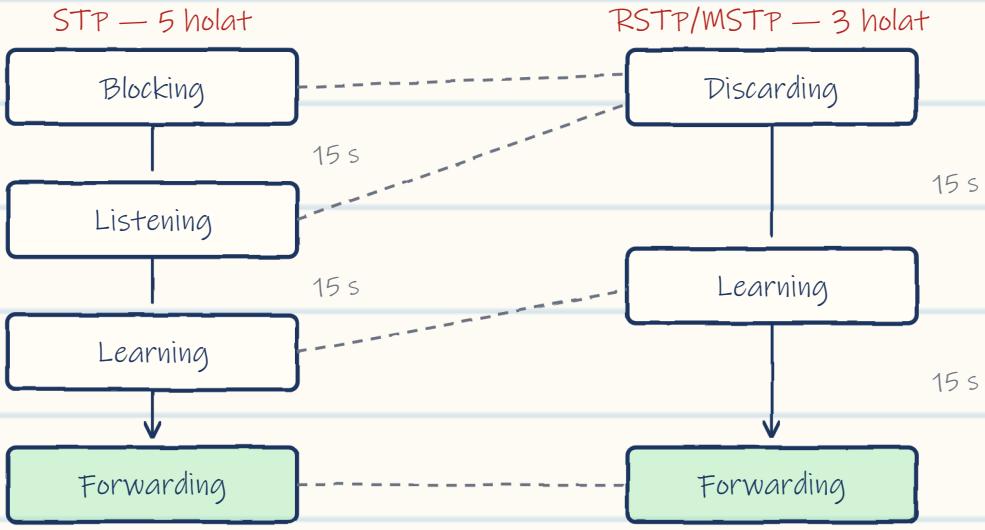
4-rasm. Narx teng → kichik bridge ID, keyin port ID

7. Port holatlari (802.1D)

Disabled	BPDU yo‘q · data yo‘q
Blocking	BPDU faqat qabul · data yo‘q
Listening	BPDU oladi va yuboradi · data yo‘q
Learning	BPDU + MAC o‘rganadi · data yo‘q
Forwarding	BPDU + MAC + data uzatadi

- Listening va Learning — oraliq holatlar

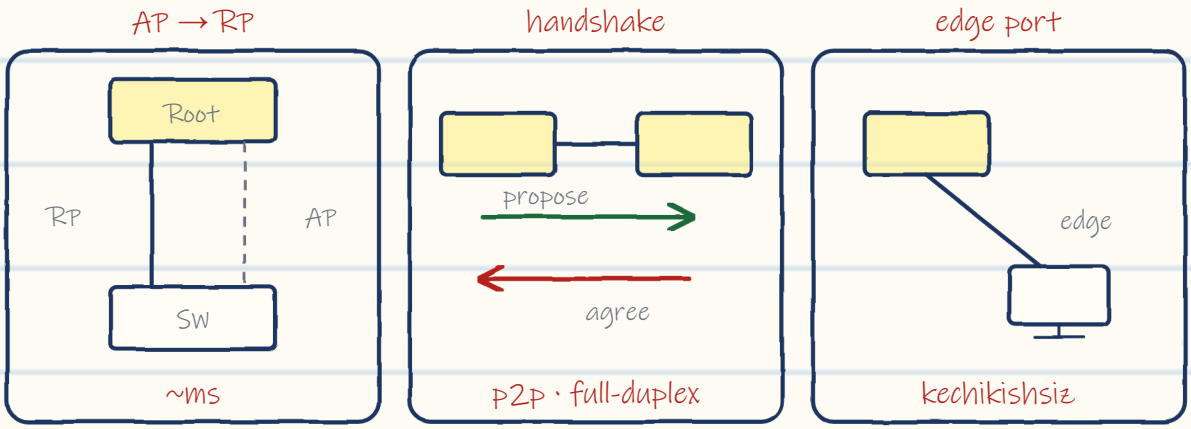
8. O‘tish va forward delay



Qoida: blocking → forwarding = 2 × 15 = 30 s. Taymerlar: Hello 2 s · MaxAge 20 s · FwDly 15 s.

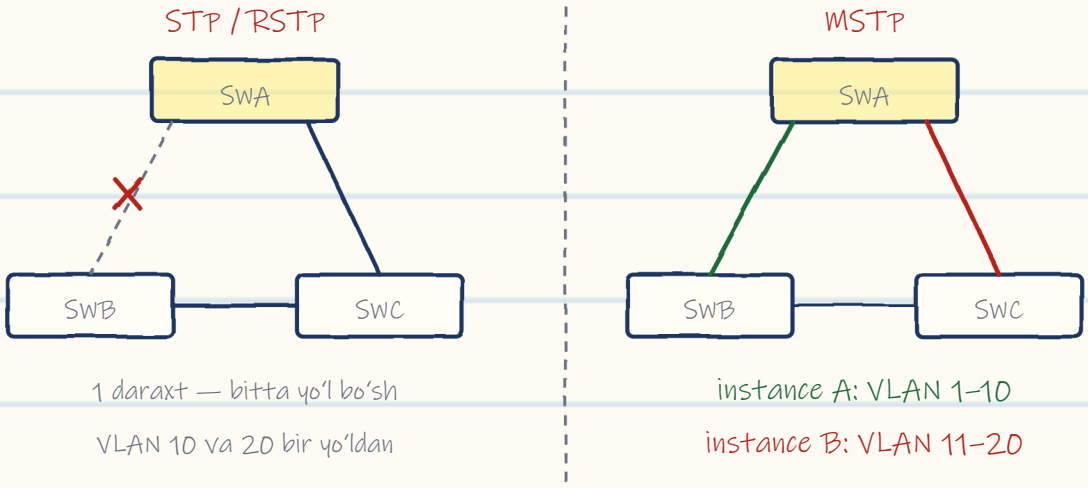
Kamchilik: topologiya tez-tez o‘zgarsa, tarmoq doim uzilib turadi.

9. RSTP — tez konvergensiya



5-rasm. RSTP = 802.1w → 802.1D-2004

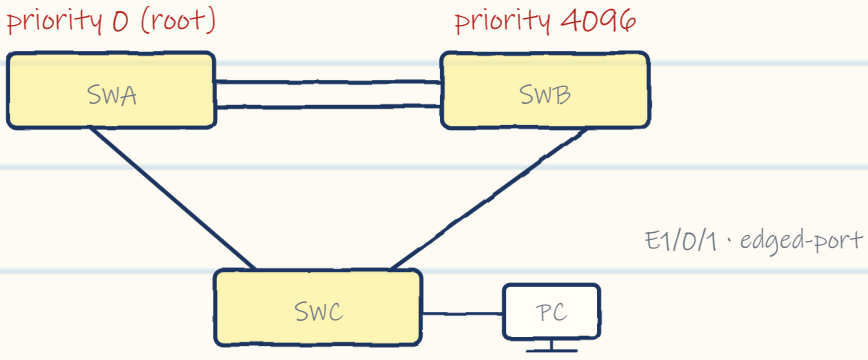
10. PVST · MSTP



	STP	RSTP	PVST	MSTP
Halqani yo'qotish	+	+	+	+
Tez konvergensiya	–	+	+	+
Yuk taqsimlash	–	–	+	+

11. Sozlash

- stp global enable — standartda o'chirilgan
- stp mode stp | rstp | pvst | mstp — standarti mstp
- stp priority 0 → root · 4096 → zaxira root
- stp edged-port · undo stp enable (port ostida)
- display stp · display stp brief



Xulosa

- STP zaxira yo'lni bloklab halqani yo'qotadi (802.1D)
- Eng kichik bridge ID → Root; narx teng bo'lsa ID, keyin port ID hal qiladi
- Rollar: DP · RP faol, AP bloklangan
- STP 5 holat, forward delay 15 s → 30 s; RSTP 3 holat
- MSTP: VLAN → instance, bir necha trunk orasida yuk taqsimlash

Tekshirish

1. Bridge ID nimadan tuziladi? — ustuvorlik + MAC
2. Forward delay va to'liq o'tish? — 15 s va 30 s
3. RSTP holatlari? — discarding, learning, forwarding
4. H3C'da 1 Gbit/s narxi? — 20